

BLINKIT SALES & OUTLET PERFORMANCE ANALYSIS

Power BI Dashboard Project | Retail Sales Analytics

Role: Data Analyst (Independent Project)

Tools Used: Power BI, DAX, Power Query (M), Excel

Dataset: Blinkit retail transaction & outlet dataset (8,523 items across 4 outlet types)

Deliverable: Interactive multi-page Power BI dashboard with cross-filtering, KPI cards, and drill-down visuals

1. Project Overview

This project analyzes retail sales performance for Blinkit, India's quick-commerce grocery platform, using a transactional dataset covering item-level sales, outlet characteristics, and location tiers. The goal was to design an interactive Power BI dashboard that enables stakeholders to explore sales performance by product category, outlet type, outlet size, location tier, and establishment year — supporting data-driven decisions on inventory allocation, outlet expansion, and category management.

2. Objective

- **Consolidate** fragmented sales data into a single interactive view for quick decision-making.
- **Identify** top- and bottom-performing product categories and outlet types.
- **Analyze** the effect of outlet size, location tier, and fat-content classification on sales volume.
- **Surface** trends in outlet establishment over time to support expansion strategy.
- **Enable** self-service filtering by outlet location type, outlet size, and item type for business users.

3. Dataset & Scope

Metric	Value
Total Sales	\$1,000,000+
Number of Items	8,523
Average Sales per Item	\$141
Average Customer Rating	3.9 / 5
Outlet Types Covered	Grocery Store, Supermarket Type 1/2/3
Location Tiers	Tier 1, Tier 2, Tier 3
Outlet Sizes	Small, Medium, High
Item Categories	16 (e.g. Fruits & Vegetables, Snacks, Household, Dairy, Frozen Foods)

4. Dashboard Design

The dashboard was structured into three functional zones for fast, guided analysis:

- **Filter Panel (left rail):** global slicers for Outlet Location Type, Outlet Size, and Item Type, allowing every visual on the page to respond dynamically.
- **KPI Summary Band:** four headline cards (Total Sales, Average Sales, Number of Items, Average Rating) for at-a-glance performance tracking.
- **Analytical Core:** a donut chart for fat-content sales split, a horizontal bar chart ranking item types by sales, a tiered bar comparison of fat content by outlet tier, a trend line for outlet establishment by year, a donut for outlet size distribution, a ranked bar list for location tier, and a detailed outlet-type performance table with five metrics (total sales, item count, average sales, item visibility, average rating).

5. Key Insights

5.1 Category Performance

- **Fruits & Vegetables and Snack Foods** are the leading categories, each contributing \$0.18M in sales — together accounting for roughly 30% of category-level revenue.
- **A steep long-tail drop-off** exists after the top five categories; categories like Seafood, Breakfast, and Starchy Foods each contribute \$0.01M–\$0.02M, signalling potential candidates for assortment rationalization.

5.2 Fat Content Segmentation

- **Regular items outsell Low Fat items** by roughly 1.8x (\$776.32K vs. \$425.36K), and this ratio holds consistently across all three location tiers — indicating the preference is structural rather than tied to a specific region.

5.3 Outlet Tier & Size Analysis

- **Tier 3 outlets** lead in total sales (\$472.13K, ~40% share), ahead of Tier 2 (\$393.15K) and Tier 1 (\$336.40K), suggesting stronger demand or outlet density in Tier 3 markets.
- **Medium-sized outlets** generate the highest sales (\$507.9K), followed by Small (\$444.79K); High-size outlets trail at \$248.99K — a signal worth investigating, since it may reflect fewer High-size outlets rather than weaker per-outlet performance.

5.4 Outlet Establishment Trend

- **A sharp spike to \$205K in 2018** stands out against an otherwise flat \$128K–\$133K range across other years — the single most notable anomaly in the dataset and a strong candidate for root-cause investigation (new outlet cohort, promotional activity, or data entry effect).

5.5 Outlet Type Performance

- **Supermarket Type 1 dominates volume**, generating \$787.5K in sales from 5,577 items — nearly five times the item count of any other outlet type.
- **Grocery Store** carries the fewest items (1,083) but matches the top average rating (4), indicating a leaner, higher-satisfaction format.
- **Supermarket Type 2 and Type 3** perform similarly in total sales (~\$130K–\$131K) despite differing average sales per item, warranting a closer per-SKU efficiency review.

6. Business Recommendations

- Prioritize inventory and shelf space for Fruits & Vegetables and Snack Foods given their consistent top-line contribution.
- Investigate the 2018 outlet establishment spike to determine whether the drivers can be replicated in future expansion planning.
- Reassess the role of High-size outlets — evaluate sales per outlet rather than raw totals to confirm whether the format is genuinely underperforming or simply under-represented.
- Use Supermarket Type 1's assortment and operating model as a benchmark for scaling other outlet types.
- Consider rationalizing long-tail categories (Seafood, Breakfast, Starchy Foods) or repositioning them with targeted promotions.

7. Skills Demonstrated

- Dashboard design & data storytelling in Power BI
- DAX measures for KPIs (Total Sales, Average Sales, Average Rating)
- Data modeling and relationship building across outlet and item dimensions
- Interactive filtering / cross-highlighting (slicers, donut, bar and line visuals)
- Business insight generation and actionable recommendation writing